

THE STRUCTURE OF MATHEMATICAL REALITY

Phase I — Chapter 1

# Mathematics Is Not About Numbers

Quiz

Self-Study Curriculum

**Before you begin**

**This quiz tests application, not memorisation. Answer in your own words.**

Do not quote the lecture back at yourself. If you find yourself writing a phrase you remember reading, stop and rephrase it from your own understanding. A short, precise answer in your own language is worth more than a long transcription.

1. A population biologist and an electrical engineer independently discover that their systems obey the same differential equation. The biologist says this is a coincidence. The engineer says it is not. Who is right, and why?

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2. Topology and geometry both study “shape.” What is the difference between them, stated in terms of what each one considers to be “the same shape”?

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3. You build a linear regression model on a biological dataset and it works well. Your colleague says, “The system must be linear.” Is this correct? What has actually been demonstrated?

4. Someone claims that “mathematics is discovered, not invented.” Someone else claims the opposite. Restate the disagreement in terms of *structure*, what would each position actually mean?

#### After the quiz

For each answer, ask yourself: “*Did I explain the **why**, or did I just state the **what**?*” If you stated a fact without explanation, go back and add the reasoning.

**Date:** \_\_\_\_\_ **Questions where I want to go deeper:**

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